

Type of HW part	Failure Mode	Distribution
AMP (Dual Differential Amplifier)	short circuits:	Birolini: Linear IC 75% functional, no shorts This is not ok! Assumption: 75%/2 sc, 75%/2 functional faults.
	sc 7-x: NC - xxx	0%
	sc 1-2: -IN - GND	2%
	sc 1-3: -IN - A1	2%
	sc 1-4: -IN - A2	2%
	sc 1-5: -IN - Out	2%
	sc 1-6: -IN - +Vs	2%
	sc 1-8: -IN - +IN	2%
	sc 2-3: GND - A1	2%
	sc 2-4: GND - A2	2%
	sc 2-5: GND - Out	2%
	sc 2-6: GND - +Vs	2%
	sc 2-8: GND - +IN	2%
	sc 3-4: A1 - A2	2%
	sc 3-5: A1 - Out	2%
	sc 3-6: A1 - +Vs	2%
	sc 3-8: A1 - +IN	2%
	sc 4-5: A2 - Out	2%
	sc 4-6: A2 - +Vs	2%
	sc 4-8: A2 - +IN	2%
	sc 5-6: Out - +Vs	2%
	sc 5-8: Out - +IN	2%
	sc 6-8: +Vs - +IN	2%
M (Motor)	Functional:	---
	wrong amplifying	9%
	internal stuck at	9%
	oscillation	9%
	offset	9%
	Break (interruption) of the coil	25%
	Short circuit of the coil	25%
	High transition resistance on the commutator	25%
	Intermittent operation	25%